

UC San Diego

JACOBS SCHOOL OF ENGINEERING

Team 2: IR Camera Human Detection System

Electrical Engineering

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What We Promised

Scope committed in the Week 1 proposal

MUST-HAVE

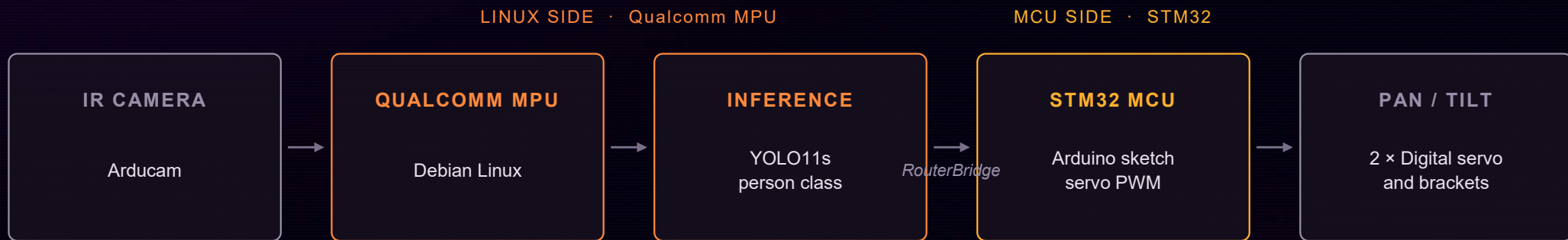
- Real-time human detection from an IR camera
- Portable enclosure housing the AUQ and the IR camera

NICE-TO-HAVE

- Live IR video feed
- Manual pan control via joystick or other external input
- Fine-grained classification beyond human / non-human (e.g. dog, cat)
- Real-world extension: restricted-area or fall detection

What We Have Done

System architecture — one board, two processors



- **IR Human Detection** Detects and labels human presence when given IR photos/videos
- **AUQ & Camera Enclosure** 3D printed separate casings for the AUQ and the camera
- **Free Camera Movement** Connected two digital servos onto a bracket that gives the camera movement across the XY-axis

What Worked

Human Detection

YOLO11s labels human presences in the IR frame with usable confidence.

AUQ & Camera Casing

Enclosures not only gave the parts protection but also allowed a smoother design process when putting the parts together

Bracket Setup

The brackets allowed easy connection with servos and a sturdy design to connect both camera and AUQ

What Did Not Work as Expected

Symptom and root cause

- **Precision vs Recall**

Models had issues with balancing between precision and recall when given other datasets to classify multiple classes besides humans
- **Power Supply**

The AUQ's 5V power pin does not supply enough power to the servos
- **Servo Jitter**

Without something to stabilize the servos, it begins to shake during movement

How Would We Solve It

Concrete fixes, mapped to the failures above

Class Reduction



Reduced the number of objects the model needed to classify

External Power Supply



Used a separate battery pack to power the servos in parallel

Stabilized the Base



3D printed a case for the bottom servo and connected it to the AUQ case

● If We Had Another Week

01

Diverse Classification

Retrain the model to recognize more objects with higher confidence

02

Design a Seamless Enclosure

Have an enclosure that can house all parts without restricting movement

03

Distance Accuracy

Train the model to have recognize distances of IR signatures with usable confidence

• Lessons Learned

• Plan More Efficiently

Reserve time and place to test camera

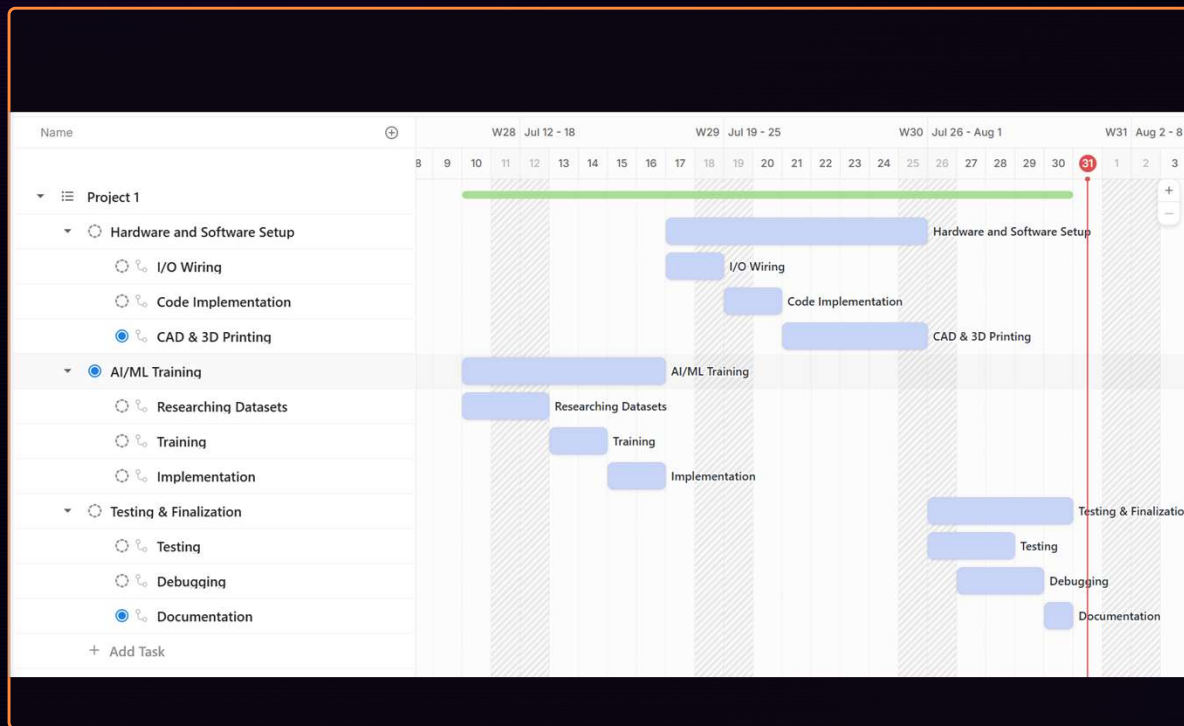
• Test Connections

YOLO11s separates a person from most static warm objects in the IR frame with usable confidence.

• Don't Short Batteries

Do not connect black and red

Updated Gantt Chart



Demonstration Video

Live run — recorded capture as fallback



Thank you

IR Camera Human Detection System · Arduino UNO Q

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